

SIMATS ENGINEERING

CO3- ASSESSMENT TOOL 3 ERROR ANALYSIS TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING COURSE CODE: ECA09

COURSE OUTCOME COVERED

CO 3: Evaluate finite word-length effects and multirate signal processing techniques, including decimation, interpolation, and polyphase decomposition. **Bloom Level mapped- BL3,BL4,BL5**

Topics Covered:

Quantization- Truncation and Rounding errors - Quantization noise - Limit cycle oscillations – Signal scaling. Parametric and Non parametric Spectral estimation. Decimation-Interpolation-Sampling rate conversion-Implementation of sampling rate conversion-Polyphase Decomposition -5G Decimation/Interpolation: For mmWave beamforming

Assessment Tool number : CO3-AT23
Assessment Tool Weightage : 40%
Assessment Tool Name : Error Analysis Task
Duration of this assessment : 60 Minutes
Marks : 25 Marks
Type of Assessment Conduction : Hybrid

- Each student will be assigned one analytical error debugging problem as a home task.
- Students must identify the implementation error(s) and analyze the root cause.
- Perform the required calculations and provide the corrected solution.
- Justify the proposed correction with appropriate DSP concepts.
- Upload the completed solution to Github and submit Git hub repository link in GCR.

Dr.

11. A signal sampled at 24 kHz is decimated by 4 without filtering. The output spectrum shows aliasing. Explain the debugging procedure. (10) (BL4) (WK4)

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Error Identification	20%	Accurately identifies all errors with clear understanding	Identifies most errors with minor omissions	Identifies limited errors with some missed	Fails to identify key errors
Root Cause Analysis	30%	Clearly explains underlying causes with strong reasoning	Explains causes with moderate clarity	Provides partial explanation with weak reasoning	Unable to identify causes
Correction	25%	Provides correct solutions with strong justification	Provides correct corrections with some justification	Provides partially correct corrections	Provides incorrect corrections
Application of Concepts	15%	Applies relevant concepts effectively	Applies concepts with minor errors	Applies concepts with guidance	Unable to apply concepts
Clarity of Presentation	10%	Presents analysis clearly with logical structure	Presents analysis with minor clarity issues	Presents analysis with poor structure	Presents unclear analysis

1. Error Identification

The main implementation error is that the signal is **decimated by 4 without applying an anti-aliasing low-pass filter before downsampling**. Decimation reduces the sampling rate from:

$$F_s = 24 \text{ kHz}$$

to:

$$F_{s,\text{new}} = \frac{24}{4} = 6 \text{ kHz}$$

Therefore, the new Nyquist frequency is:

$$F_N = \frac{6}{2} = 3 \text{ kHz}$$

Any frequency components above **3 kHz** in the original signal can cause aliasing after decimation.

2. Root Cause Analysis

The root cause is the **absence of an anti-aliasing filter before downsampling**.

In decimation, samples are directly discarded by a factor of 4. If the original signal contains frequency components above 3 kHz, those components are not removed before the sampling rate is reduced. They overlap with the valid frequency range after downsampling and appear as incorrect frequency components in the output spectrum. Thus, the observed aliasing is caused by **improper implementation of the decimation process**

3. Required Correction

The correct decimation procedure is:

Input signal → Anti-aliasing Low-pass Filter → Downsample by 4 → Output

The low-pass filter should restrict the signal bandwidth to approximately **3 kHz or below** before downsampling.

Then the filtered signal can be downsampled by taking every fourth sample:

$$y[n] = x_f[4n]$$

where $x_f[n]$ is the filtered signal.

The corrected sampling rate becomes:

$$F_{s,\text{out}} = 6 \text{ kHz}$$

and its Nyquist frequency is:

3 kHz

This prevents frequency components above the new Nyquist limit from producing aliasing.

4. Application of DSP Concepts

The debugging procedure should be:

1. **Check the original sampling frequency:** 24 kHz.
2. **Check the decimation factor:** $M = 4$.
3. **Calculate the new sampling frequency:** $24/4 = 6$ kHz.
4. **Calculate the new Nyquist frequency:** $6/2 = 3$ kHz.
5. **Inspect the input spectrum** for components above 3 kHz.
6. **Check whether an anti-aliasing low-pass filter is present.**
7. If the filter is missing, **add the filter before decimation.**
8. Perform downsampling only after filtering.
9. Compare the output spectrum before and after correction.

5. Final Debugging Conclusion

The aliasing occurs because the signal was **downsampled directly without anti-aliasing filtering**. The sampling rate decreases from **24 kHz to 6 kHz**, reducing the Nyquist frequency to **3 kHz**. Therefore, an appropriate low-pass anti-aliasing filter must be applied before decimation. After filtering and then downsampling by 4, the output spectrum should be free from aliasing.